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Midterm 2

Unit 1

1 a) $\vec{V}$: up, $\vec{F}$: Left $\rightarrow$ $\vec{B}$ = into page

b) $\vec{V}$: down, $\vec{F}$: Into page $\rightarrow$ $\vec{B}$ = Left

c) $\vec{V}$ Left, $\vec{F}$: up $\rightarrow$ $\vec{B}$ = out of page

2. a) Coil 1 $\rightarrow$ Switch closed: Clockwise current
After long time: No current change
Switch opened: Counterclockwise current

b) Coil 2 $\rightarrow$ switch closed: Clockwise current
After long time: No current change
Switch opened: Counterclockwise current

c) Coil 3 $\rightarrow$ switch closed: Clockwise current
After long time: No current change
Switch opened: Counterclockwise current

3. a) $\Phi_B = B \cdot A \cdot \cos\theta$ $A = \pi\left(\frac{d}{2}\right)^2 \rightarrow \pi\left(\frac{.0220}{2}\right)^2 \rightarrow 3.801 \cdot 10^{-4}\ m^2$ $2.2cm \rightarrow .022m$

$B = 2T$ $3.801 \cdot 10^{-4} \cdot 2 \rightarrow \Phi_B = 7.602 \cdot 10^{-4}\ wb$

$EMF = \frac{\Delta\Phi_B}{\Delta t} \rightarrow \frac{7.602 \cdot 10^{-4}}{.250} \rightarrow 3.041 \cdot 10^{-3}\ V \rightarrow 3.04\ mV$

b) $I = \frac{EMF}{R} \rightarrow \frac{3.041 \cdot 10^{-3}}{.0100} \rightarrow .304\ A$

$R = .0100$

c) $P = I^2 R$

$.304^2(.0100) \rightarrow .0924 \cdot .0100 \rightarrow 9.24 \cdot 10^{-4}\ W$

4. $\mathcal{E}_{max} = NBA\omega$

a) $N = \frac{\mathcal{E}_{max}}{BA\omega}$

$\omega = 1875$ rad/s

$\mathcal{E}_{max} = 18V$

$B = .640T$

$A = 1cm \times 3\,cm \rightarrow .01 \cdot .03 = .0003$ or $3 \cdot 10^{-4} m^2$

$N = \frac{18}{.64(3 \cdot 10^{-4})(1875)} \rightarrow \frac{18}{.36} = 50$ turns

b) $T = \frac{2\pi}{\omega} \longrightarrow \frac{2\pi}{1875} = .00335103\,22$ $~~3~~.35 \cdot 10^{-3}$ s or 3.35 ms

5. $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$

$V_p = 240$ V

$V_s = 120$ V

a) $\frac{240}{120} = \frac{2}{1}$ 2 turns of primary coil per 1 turn of secondary

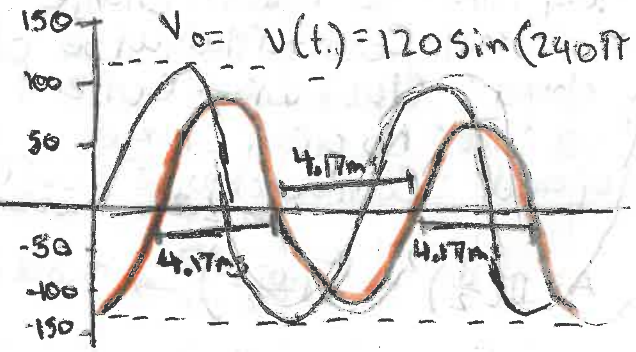
b) $\frac{N_p}{N_s} = \frac{I_s}{I_p} = 2 \rightarrow \frac{2}{1} \rightarrow \frac{1}{2}$ input current $= \frac{1}{2}$ of output current

6. a)

$V_0 = 120$ V

$\omega = 240\pi$

$\frac{240\pi}{2\pi} = 120$ Hz



b) $\mathcal{E}_s(t) = -N\frac{d\Phi_B}{dt}$

↓

cos wave

(shift 90° from $v(t)$)

c) $v(t) = 120\sin(240\pi t) = 0$

$\sin(240\pi t) = 0 \rightarrow 240\pi t = n\pi \rightarrow t = \frac{n}{240}$

$\sin\theta = 0$ when it is multiple of π $\theta = \pi(n)$

counter

$\frac{240\pi t}{\pi} = \frac{\pi n}{\pi} \rightarrow \frac{240t}{240} = \frac{n}{240}$

$\frac{n}{240} = t$

$\frac{1,2,3}{240}$ ~~only~~ $\frac{1}{240}$ s $\rightarrow 4.17$ ms. $V = 0$

7. $I = 0.1$ A

$L = 200$ mH $\rightarrow 2 \cdot 10^{-3}$ H

$\mathcal{E} = 500$V

$\mathcal{E} = L\frac{\Delta I}{\Delta t}$

$\frac{L\Delta I}{\mathcal{E}} = \Delta t$

$\frac{2 \cdot 10^{-3} \cdot .1}{500} \rightarrow \frac{2 \cdot 10^{-4}}{5\cancel{00}} \rightarrow \frac{2 \cdot 10^{-6}}{5} \rightarrow .4 \cdot 10^{-6} \rightarrow 4 \cdot 10^{-7}$ s

Unit 1

8) a) $\varepsilon = L\frac{\Delta I}{\Delta t}$ $L = 25\ H$ $I = 100A$ $\Delta t = 80ms \rightarrow .08s$

$\varepsilon = 25 \cdot \frac{100}{.08}$

$\varepsilon = 31250\ V$

b) $E = \frac{1}{2}LI^2$

$E = \frac{1}{2}(25)(100)^2$

$12.5 \cdot 10^4$

$1.25 \cdot 10^5\ J$

c) $P = \frac{\text{Energy}}{T}$ $\frac{1.25 \cdot 10^5}{.08} = 1562500$

$1.5625 \cdot 10^6\ W$

d) $L = \mu_0 \frac{N^2 A}{l}$ $\mu_0 = 4\pi \cdot 10^{-7}$

$25 = 4\pi \cdot 10^{-7}$

I choose: 10,000 turns $(10^4) = N$
$.1\ m^2 = A\ (10^{-1})$ $\rightarrow$ $25 = (4\pi \cdot 10^{-7}) \frac{(10^4)^2 \cdot 10^{-1}}{l}$

$.5026548246 \leftarrow = \frac{4\pi}{25} = l \leftarrow l = \frac{4\pi \cdot 10^{-7} \cdot 10^8 \cdot 10^{-1}}{25}$

Selenoid length = .5 m^2, 10,000 turns, .1m^2 cross sectional area

e) $X_L = 2\pi f L$ 10 KHz $\rightarrow$ 10000 Hz
$2\pi \cdot 10000 \cdot 25$ $500000\pi \rightarrow 1.57 \cdot 10^6\ \Omega$

9. a) $\tau = \frac{L}{R} \rightarrow \tau R = L$ $20 \cdot 10^{-9}s \cdot 5 \cdot 10^6\ \Omega$
$10^2 \cdot 10^{-9} \cdot 10^6 \cdot H \rightarrow .1\ H$

$\tau = 20\ ns \rightarrow 20 \cdot 10^{-9}\ s$
$R = 5.00\ M\Omega\ 5 \cdot 10^6\ \Omega$

b) $\tau = \frac{L}{R} \rightarrow R = \frac{L}{\tau}$

$\tau = 1\ ns \rightarrow 1 \cdot 10^{-9}\ ns \rightarrow R = \frac{10^{-1}}{10^{-9}} \rightarrow 10^8\ \Omega$
$L = .1\ H\ (10^{-1})$

c) $I(t) = I_0 e^{-t/\tau}$ $I \frac{3 \cdot 10^{-9}}{20 \cdot 10^{-9}}$ $I(t) = I_0 \cdot e^{-.15} \rightarrow \frac{I(t)}{I(0)} = e^{-.15}$

$.8607079764$

$\approx 86.07\%$

$t = 3ns \rightarrow 3 \cdot 10^{-9}s$
$\tau = 20\ ns \rightarrow 20 \cdot 10^{-9}s$

10. a) $V_{in} = V_R + V_C = jR + V_{out}$

$V_{in} = \frac{dq}{dt}R + V_{out}$ $\frac{dq}{dt}$

b) $V_{out} = V_C = \frac{q}{C}$

c) $\frac{V_{out}}{V_{in}} = \frac{Z_C}{Z_R + Z_C}$ $Z_C = \frac{1}{j\omega C}$ $Z_R = R$

$\frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \cdot \frac{j\omega C}{j\omega C}$ $\frac{1}{R(j\omega C)+1} \rightarrow \frac{1}{Rj\omega C+1}$

d) $f = 100(2\pi RC)^{-1}$

$\omega = 2\pi f = 2\pi \cdot 100 \cdot \frac{1}{2\pi RC} = \frac{100}{RC} \rightarrow \frac{V_{out}}{V_{in}} = \left|\frac{1}{1+j(100)}\right| = \frac{1}{\sqrt{1+100^2}}$

$\rightarrow \frac{1}{\sqrt{10001}} = .0099995$ strongly reduced

$f = .1(2\pi RC)^{-1}$

$\omega = 2\pi f = 2\pi \cdot .1 \cdot \frac{1}{2\pi RC} = \frac{.1}{RC} \rightarrow \frac{V_{out}}{V_{in}} \rightarrow \left|\frac{1}{1+j(.1)}\right| = \frac{1}{\sqrt{1+.1^2}}$

$\frac{1}{\sqrt{1.01}} \rightarrow 0.995$ → Allmost unchanged

11. a) $f_0 = \frac{1}{2\pi\sqrt{LC}}$ $\rightarrow \frac{1}{2\pi\sqrt{10^{-6} \cdot 10^{-2}}} \rightarrow \frac{1}{2\pi(10^{-8})^{\frac{1}{2}}} \rightarrow \frac{1}{2\pi \cdot 10^{-4}} \rightarrow \frac{10^4}{2\pi} \rightarrow 1591.549431$

$1591.55\ Hz$

$R = .1\ k\Omega = 100\ \Omega$

$C = 1\ \mu F = 1 \cdot 10^{-6} F$

$L = 10\ mH = 10 \cdot 10^{-3} H$

b) $Z = \sqrt{R^2 + (X_L - X_C)^2}$ $f = .1$ $f_0 = 159.2\ Hz$

$X_L = 2\pi f L$ $X_L = 2\pi \cdot 159.2 \cdot .01 = 10\ \Omega$

$X_C = \frac{1}{2\pi f C}$ $X_C = \frac{1}{2\pi \cdot 159.2 \cdot 10^{-6}} = \approx 1000\ \Omega$

$Z = \sqrt{100^2 + (10-1000)^2} \rightarrow \sqrt{10^4 + 990^2} \rightarrow \sqrt{970100}$

$994.9\ \Omega$

$f = 10 f_0 = 15920\ Hz$

$X_L = 2\pi \cdot 15920 \cdot .01 \approx 1000\ \Omega$

$X_C = \frac{1}{2\pi \cdot 15920 \cdot 10^{-6}} \rightarrow 10\ \Omega$

$Z = \sqrt{10^4 + (1000-10)^2} \rightarrow \sqrt{10^4 + 990^2} \rightarrow \sqrt{970100}$

$994.9\ \Omega$

$994.9\ \Omega$

12. a) $I_{rms} = \frac{V_{rms}}{Z}$

$f = .1 f_0 = 159.2\ Hz$

$Z = 994.9\ \Omega$

$\frac{120}{994.9} = .1206\ A$

$f = 10 f_0 = 15920\ Hz$

$Z = 994.9\ \Omega$

$\frac{120}{949.9} = .1206\ A$

b) $P_{rms} = V_{rms} I_{rms} \cos\phi$

$f = .1 f_0$ $X_L - X_C = 10 - 1000 = -990$

$\phi = \tan^{-1}(-990/100) \approx -84.23°$

$\cos\phi = \cos(84.23) \approx .100535365$

$P_{rms} = 120 \cdot .1206 \cdot .100535365$

$P_{rms} = 1.454947803 = 1.455\ W$

$f = 10 f_0$ $X_L = X_C = 1000 - 10 = 990$

$\phi = \tan^{-1}(990/100) \approx 84.23°$

$\cos\phi = .100535365$

$P_{rms} = 120 \cdot .1206 \cdot .100535365$

$P_{rms} = 1.455\ W$

Unit 1

13. $f_c = 1.4$ MHz → 1400 kHz Carrier frequency: $f_c = 1.4$ MHz
$f_m = 10$ kHz → .01 MHz

Upper Sideband:
$f_c + f_m = 1400 + 10 = 1410$ kHz
or $1.4 + .01 = 1.41$ MHz

lower side band:
$f_c - f_m = 1.4$ MHz $- .01$ MHz → 1.39 MHz
or 1400 kHz − 10 kHz = 1300 kHz

Unit 2

1. a) $B(2\pi r) = \mu_0 I_{displacement}$ $r = 1cm \rightarrow .01m$

$\mu_0 = 4\pi \cdot 10^{-7}\ T \cdot m/A$

$I = \frac{\Delta Q}{\Delta t}$ $\Delta Q = 200nC \rightarrow 200 \cdot 10^{-9} C$

$\Delta t = 2\ \mu s = 2 \cdot 10^{-6} s$

$\frac{200 \cdot 10^{-9}}{2 \cdot 10^{-6}} \rightarrow I_{displacement} = .1\ A$

$B = \frac{\mu_0 I}{2\pi r}$

$B = \frac{(4\pi \cdot 10^{-7})(.1)}{2\pi(.01)} \rightarrow 2 \cdot 10^{-6}\ T$

b) Displacement Current Responsible

.1 A

2. a) $\Delta t = 10\mu s = 10 \cdot 10^{-6} \rightarrow 10^{-5}\ s$ $d = \frac{c\Delta t}{2}$

c (speed of light) $= 3 \cdot 10^{8}\ m/s$

$d = \frac{3 \cdot 10^{8} \cdot 10^{-5}}{2} \rightarrow \frac{3 \cdot 10^{3}}{2}$ $1.5 \cdot 10^{3}$ 1500 m

b) $R = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2$ $1.78 = n_2$ $\left|\frac{1 - 1.78}{1 + 1.78}\right|^2 \rightarrow .0787$

7.87%

3. a) $I = \frac{P}{A}$ → Power, → Area $P = 1kW \rightarrow 1000 (10^3 w)$

$A = 10\ cm^2 \rightarrow 10^{-3}\ m^2$

$I = \frac{10^3\ W}{10^{-3} m^2} \rightarrow 10^6\ w/m^2$

b) $I_2 = I_1 \cdot \left(\frac{r_1}{r_2}\right)^2 \rightarrow I_2 = 10^6 \cdot \left(\frac{1}{2}\right)^2 \rightarrow 10^6 \cdot \frac{1}{4} \rightarrow 2.5 \cdot 10^5\ w/m^2$

$I_1 = 10^6\ W/m^2$ $W/m^2 \cdot \frac{m}{m}$

$r_1 = 1\ m$

$r_2 = 2m$

$2.5 \cdot 10^5\ w/m^2$

$n_1 = n_3$

4. a) $n_1 \sin\theta_1 = n_2 \sin\theta_2$ + $n_2 \sin\theta_2 = \cancel{n_3} \sin\theta_3$
$\downarrow$ n_1

$\frac{n_1}{n_2} \sin\theta_1 = \sin\theta_2$

$n_2 \left(\frac{n_1}{n_2} \sin\theta_1\right) = n_1 \sin\theta_3$

$\cancel{n_1} \sin\theta_1 = \cancel{n_1} \sin\theta_3$

$\sin\theta = \sin\theta_3$

b) $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$ $\quad \frac{1}{f} - \frac{1}{d_o} = \frac{1}{d_i}$ $\quad \frac{1}{d_i} = \frac{1}{12} - \frac{1}{8} \rightarrow \frac{2}{24} - \frac{3}{24}$ $\quad \frac{1}{d_i} = -\frac{1}{24}$ $\quad d_i = -24\text{cm}$

$M = \left|\frac{d_i}{d_o}\right| \rightarrow M = \left|\frac{-24}{8}\right|$ $\quad M = 3$

5. $\sigma = 200$ BARNS $\rightarrow 2 \cdot 10^{-22}\text{cm}^2$
$d = 1\text{cm}$
$\rho = 11.34\ \text{g/cm}^3$
$N_A = 6.022 \cdot 10^{23}/\text{mol}$
$A = 207.2\ \text{g/mol}$

$\rightarrow n = \frac{\rho N_A}{A}$ $\quad \frac{11.34 \cdot 6.022 \cdot 10^{23}}{207.2} \rightarrow 3.29 \cdot 10^{22}$

$f = e^{-n\sigma d} \rightarrow e^{-3.296 \cdot \cancel{10^{22}} \cdot 2 \cdot \cancel{10^{-22}} \cdot \cancel{1}} \rightarrow e^{-3.296 \cdot 2}$

$\downarrow$

$e^{-6.596} \rightarrow .0013712946$

$.137\%$ of x-rays pass through

6. a) $E = 250\text{mJ} \rightarrow .25\text{J}$ $\quad .01\ \text{J/kg}$ per rad
$m = 60\text{kg}$

Dose $= \frac{E}{m} \rightarrow \frac{.250\text{J}}{60\text{kg}} = 0.004166\overline{7}\ \text{J/kg} \xrightarrow{\times 100} .41667 \approx .417$ rad = dose

b) $m = 2\text{kg}$
$E = 250\text{mJ} \rightarrow .25\text{J}$ $\quad \frac{E}{m} \rightarrow \frac{.250\text{J}}{2\text{kg}} \rightarrow .125\ \text{J/kg} \xrightarrow{\times 100} 12.5$ rad = dose

c) RBE = 1
1 rad = .01 Gy
1 Gy $\cdot$ RBE = SV

$12.5\ \text{rad} \cdot .01 \frac{\text{Gy}}{\text{rad}} = .125\ \text{GY} \longrightarrow .125 \cdot 1 = .125$

$.125$ SV

d) No, this dose is considered relatively small (safe)